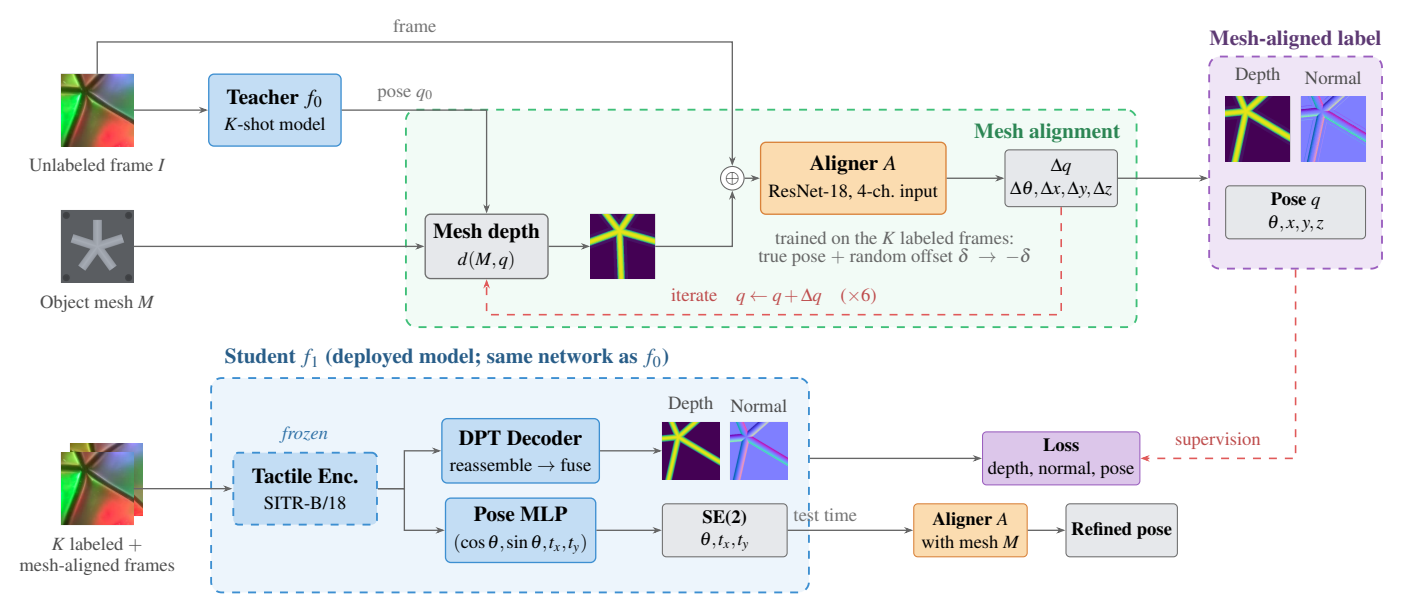

frame
Unlabeled frame $I$
Teacher $f_0$
$K$-shot model
pose $q_0$
Object mesh $M$
Mesh depth
$d(M,q)$
Mesh alignment
Aligner $A$
ResNet-18, 4-ch. input
$\Delta q$
$\Delta\theta, \Delta x, \Delta y, \Delta z$
trained on the $K$ labeled frames:
true pose + random offset $\delta \rightarrow -\delta$
iterate $q \leftarrow q + \Delta q$ $(\times 6)$
Mesh-aligned label
Depth
Normal
Pose $q$
$\theta, x, y, z$
Student $f_1$ (deployed model; same network as $f_0$)
$K$ labeled +
mesh-aligned frames
frozen
Tactile Enc.
SITR-B/18
DPT Decoder
reassemble → fuse
Pose MLP
$(\cos\theta, \sin\theta, t_x, t_y)$
Depth
Normal
SE(2)
$\theta, t_x, t_y$
Loss
depth, normal, pose
supervision
test time
Aligner $A$
with mesh $M$
Refined pose